

Project Title

Self-Supervised Learning for Abnormality Detection in Musculoskeletal Radiographs Using the MURA Dataset

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Group 7

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Abstract

Musculoskeletal disorders are a leading cause of long-term pain and disability worldwide, necessitating accurate and scalable diagnostic tools. This study investigates the use of self-supervised learning (SSL) frameworks—SimCLR and MoCo—for abnormality detection in musculoskeletal radiographs, leveraging the MURA dataset, a large-scale collection of labeled X-ray studies. Unlike traditional supervised methods that require extensive annotated data, SSL models can learn robust image representations from unlabeled data by leveraging contrastive learning techniques. The proposed models were pretrained on the MURA dataset using SimCLR and MoCo, followed by fine-tuning for binary classification (normal vs. abnormal). Experimental results demonstrate that both models achieved competitive accuracy, with SimCLR slightly outperforming MoCo in generalization, while MoCo attained a higher AUROC (0.79). Grad-CAM visualizations further confirmed that the models focus on clinically relevant features. This work highlights the potential of SSL in developing label-efficient and interpretable diagnostic systems for radiographic abnormality detection, paving the way for broader adoption in clinical practice.

1.Introduction:

Musculoskeletal disorders affect more than 1.7 billion people globally and represent a leading cause of long-term pain and physical disability. Accurate interpretation of musculoskeletal radiographs is crucial for diagnosing conditions such as fractures, joint diseases, and post-operative complications. However, the interpretation task can be time-consuming and error-prone, especially under high workloads and limited availability of expert radiologists. As imaging demands continue to rise, the development of robust, automated diagnostic tools has become increasingly important.

This study focuses on the MURA dataset, a large-scale collection of 40,561 musculoskeletal radiographs across 14,863 patient studies, each labeled by radiologists as either *normal* or *abnormal*. The dataset covers seven upper-extremity anatomical categories: elbow, finger, hand, humerus, forearm, shoulder, and wrist. While MURA is one of the largest labeled datasets for this domain, the labeling process is still resource-intensive, highlighting a broader issue in medical AI: the scarcity and cost of high-quality labeled data.

To overcome this limitation, self-supervised learning (SSL) offers a compelling alternative by allowing models to learn meaningful visual representations from unlabeled data. SSL approaches have shown state-of-the-art performance in various computer vision tasks by pretraining models using automatically generated pretext tasks, which are later fine-tuned using minimal labeled examples. By leveraging SSL, we aim to reduce reliance on manual annotation while maintaining or improving downstream classification performance.

In this work, we apply three prominent SSL frameworks—SimCLR, MoCo, and BYOL—to pretrain feature encoders on the MURA dataset. These encoders are then used to generate embeddings for a downstream binary classification task (normal vs. abnormal). Our objective is to compare the performance of these SSL-based models against a standard supervised baseline and to assess their potential in building scalable, label-efficient diagnostic systems for musculoskeletal radiographs.

2.Literature review:

The application of deep learning for musculoskeletal abnormality detection has gained momentum with the availability of large annotated datasets. The MURA dataset, released by Rajpurkar et al. (2018), is one

of the most comprehensive publicly available datasets in this domain, containing over 40,000 radiographs labeled as normal or abnormal by radiologists. A 169-layer DenseNet trained on MURA achieved an AUROC of 0.929, with sensitivity and specificity comparable to expert radiologists in certain body parts like wrist and finger, though lagging in others such as the humerus and shoulder [1].

Further research, such as that by Solovoya and Solovyov (2020), has demonstrated the continued improvement of computer-aided diagnosis (CAD) systems using CNNs. Their DenseNet-169 based model achieved kappa scores of 0.942 on wrist and 0.862 on hand radiographs—surpassing previous benchmarks. They emphasized that despite strong results, limitations remain due to data diversity, image resolution, and lack of fine-grained labeling. The authors argue for future work focused on higher-resolution images, more granular annotations, and integration into clinical workflows to combat radiologist fatigue and improve diagnostic speed [2].

Deep learning has become a key method for detecting abnormalities in musculoskeletal radiographs, especially with the availability of the large-scale MURA dataset. Supervised models such as ResNet and DenseNet have achieved high diagnostic accuracy but require extensive labeled data (Duan et al., 2024). To improve robustness and generalization, He et al. (2021) introduced a calibrated ensemble model combining ConvNet, ResNet, and DenseNet. Their method outperformed individual models and even expert radiologists in several anatomical regions, achieving an AUC of up to 0.97. These developments demonstrate the effectiveness of deep models while also emphasizing the need for label-efficient approaches—such as self-supervised learning—to further advance radiograph-based abnormality detection [3,4].

To further reduce dependence on labeled data, recent work has explored self-supervised and semi-supervised methods. Spahr et al. (2021) proposed a self-taught anomaly detection framework that uses contrastive pretext tasks to learn robust representations from unlabeled X-rays. Their method achieved higher AUC scores than both supervised and previous semi-supervised baselines, demonstrating better generalization and faster convergence for anomaly detection tasks on the MURA dataset [5].

The application of segmentation-based deep learning models in musculoskeletal analysis has also gained attention. A study by Chakraborty et al. (2021) employed a U-Net-based CNN for bone fracture segmentation using the MURA v1.1 dataset. Their model achieved a training Dice coefficient of 95.95% and a validation Dice coefficient of 90.29%, demonstrating high segmentation accuracy. While promising, the study notes the need for expanding the dataset to cover more fracture types and anatomical diversity, as well as exploring transformer-based models and clinical integration for enhanced diagnostic utility [6].

In efforts to develop trustworthy deep learning frameworks for shoulder abnormality detection, Mahbub et al. (2024) proposed a calibrated ensemble of ConvNet, ResNet, and DenseNet models. Their approach outperformed individual models and even expert radiologists across several evaluation metrics, including AUC and Cohen's kappa. The study also incorporated Class Activation Mapping (CAM) to improve interpretability. The authors identified integration of unsupervised clustering algorithms as a key direction for future enhancement, to uncover latent subgroups and further improve diagnostic precision [7].

Addressing the limitations of labeled data, Rahman et al. (2023) introduced a deep transfer learning (TL) strategy with enhanced feature fusion for abnormality detection in humerus and wrist X-rays. Using a two-phase TL process with InceptionResNetV2 and Xception, they implemented feature fusion followed by classification via traditional machine learning classifiers. Visualization tools such as Grad-CAM and LIME validated model decisions. Their approach demonstrated improved diagnostic performance, particularly when adapting pre-trained models to the medical imaging domain. Future research may involve expanding TL sources to include other modalities like MRI and CT to increase generalizability[8].

A hybrid deep learning approach was explored by Shibly et al. (2022), who developed the ComDNet-512 architecture to detect finger abnormalities from MURA X-rays. The model follows a three-stage pipeline including compression, CNN training, and progressive resizing to improve multi-scale feature learning. The method yielded promising results, and the authors suggested extending the model to additional anatomical regions like wrists and elbows, and integrating Hadoop for large-scale dataset handling [9].

Hasan et al. (2023) proposed an ensemble framework combining DenseNet169, MobileNet, and XceptionNet for abnormality classification in musculoskeletal radiographs. Each model independently processes input data, with predictions combined via weighted averaging. This ensemble strategy outperformed individual models in accuracy and AUC. While the paper does not explicitly outline future work, the authors imply opportunities in expanding the ensemble with other architectures and applying the methodology to diverse imaging datasets [10].

To capture both local and global features in radiographs, Khan et al. (2023) introduced MSCNN-GCN, a hybrid architecture combining Multi-Scale Convolutional Neural Networks with Graph Convolutional Networks. The MSCNN component extracts fine-grained local details, while the GCN captures contextual relationships across image regions. The combined output significantly improved abnormality classification. Future directions include extending the framework to more complex tasks like lesion segmentation in CT images of pulmonary or cardiovascular structures [11].

Ahmed et al. (2023) proposed a two-stage unsupervised deep clustering model for both classification and anomaly detection in MURA X-rays. The method uses ResNet-50 for feature extraction, followed by k-means clustering to generate pseudo-labels, enabling iterative refinement without human annotations. The framework showed strong performance in categorizing bone types and detecting abnormalities. However, the authors acknowledged dataset limitations and emphasized the need for generalizable models across diverse imaging conditions and anatomical regions [12].

In a comparative study, Bhuiyan et al. (2023) evaluated four deep learning models—YOLOv3, YOLOv7, EfficientDet-D1, and CenterNet—for bone joint localization in upper limb radiographs. YOLOv7 achieved the highest performance with an mAP of 48.3. The study emphasized the potential of larger models such as YOLOv7-E6E and the importance of hyperparameter tuning and dataset diversification to improve localization accuracy [13].

A lightweight model named FusionNet was proposed by Zhou et al. (2023) for musculoskeletal abnormality detection, combining an enhanced EfficientNet-B2 (with coordinate attention and modified bottlenecks) and MobileNetV2. This fusion aimed to retain high accuracy while reducing model complexity. Although future directions were not explicitly discussed, the model shows promise for real-time clinical applications and broader deployment in resource-constrained settings [14].

2.1. Related Works

Ref No	Method	Dataset	Best Model	Accuracy	Limitations
[1]	The model uses a 169-layer convolutional neural network for predictions. It predicts the probability of abnormality for each view. The overall probability is computed as the arithmetic mean of individual probabilities.	The dataset consists of 40,561 musculoskeletal radiographs from 14,863 studies. Each study is labeled as normal or abnormal by radiologists. Additional labels were collected from six board-certified Stanford radiologists.	The model achieved an AUROC of 0.929, indicating strong performance in detecting abnormalities. On finger and wrist studies, model performance matched the best radiologist performance.	The model achieves an area under the ROC curve (AUROC) of 0.929, with a sensitivity of 0.815 and specificity of 0.887.	There is a need for more studies to evaluate optimal integration of the model in clinical settings. Radiologist fatigue significantly impacts diagnostic accuracy, affecting performance towards the end of the workday. The model's inability to detect abnormalities as effectively as radiologists indicates a limitation in its current capabilities.
[2]	A 169-layer DenseNet CNN was used for abnormality detection in musculoskeletal radiographs. The model was initialized with ImageNet-pretrained weights and trained using a batch size of 8, over 35–50 epochs. Performance was evaluated using accuracy and Cohen's kappa	MURA	The model used for detecting bone abnormalities is based on a 169-layer Convolutional Neural Network (CNN) architecture, specifically utilizing DenseNet.	The model achieved Cohen's kappa scores of 0.942 (wrist), 0.862 (hand), and 0.735 (shoulder).	The study is limited by inconsistent image quality, lack of universal imaging standards, and the need for more advanced neural network architectures to improve diagnostic accuracy.
[3]	The methodology involves a hybrid model HRD (Human-Resnet50-Dense net121) that utilizes deep learning and human participation to classify X-ray images for diagnosing musculoskeletal diseases	MURA	The best model identified in the research is the hybrid model HRD (Human-Resnet50-Densenet121)	The hybrid model HRD achieved a peak accuracy of 88.81%, surpassing the accuracy of individual models by more than 4% across all types of X-ray images	The study showed less impact among senior physicians specializing in musculoskeletal disease diagnosis. Crucial environmental factors were overlooked, suggesting areas for future exploration.

Ref No	Method	Dataset	Best Model	Accuracy	Limitations
[4]	The study introduces a calibrated ensemble of deep learners, leveraging the strengths of ConvNet, ResNet, and DenseNet for detecting abnormalities in musculoskeletal radiographs	MURA	The best model identified in the study is the calibrated ensemble approach, which consistently outperforms the other models across five evaluation metrics and even surpasses expert radiologists in specific anatomical regions	The overall performance of the proposed model achieved an accuracy of 0.87, while the best individual model, ResNet, reached an accuracy of 0.92 in detecting abnormalities in the Humerus region	The approach requires a pre-defined partition of data subgroups, which may not be applicable to all datasets].Not all datasets exhibit straightforward distinctive data clusters, limiting the method's generalizability
[5]	The proposed method is a self-taught anomaly detection approach that utilizes a universal semi-supervised framework to generalize well for detecting unseen anomalies without prior domain knowledge	MURA	The research paper does not explicitly identify a "best model" but highlights that their method outperforms baselines across unsupervised and self-supervised anomaly detection settings on the MURA dataset	The research indicates that the proposed method achieves a validation AUC of 78% after a few epochs, outperforming the AE initialization which only reaches an AUC of around 75% after 100 epochs	Supervised models require a large number of labeled samples, which are expensive to acquire. They suffer from severe class imbalance issues due to rare anomalies
[6]	The model uses a U-Net CNN for bone fracture segmentation in musculoskeletal X-rays.	MURA	U-Net architecture	Dice coefficient: 95.95% (training), 90.29% (validation)	Future work includes expanding dataset diversity, using advanced architectures (e.g., transformers), and clinical validation.
[7]	A calibrated ensemble of ConvNet, ResNet, and DenseNet for abnormality detection in shoulder radiographs.	MURA	Calibrated ensemble model	High AUC and Cohen's kappa; outperformed expert radiologists	Limited integration of unsupervised clustering; further exploration of diverse deep learning architectures is needed.
[8]	Two-phase deep transfer learning with enhanced feature fusion and classical ML classifiers.	MURA	InceptionResNetV2 + Xception with feature fusion	Improved diagnostic accuracy, especially in wrist and humerus	Domain mismatch, reliance on pre-trained models; needs expansion to other imaging modalities (e.g., MRI, CT).

Ref No	Method	Dataset	Best Model	Accuracy	Limitations
[9]	A hybrid CNN model (ComDNet-512) with progressive image resizing for finger abnormality detection.	MURA	ComDNet-512	Accuracy not explicitly stated; robust across image resolutions	Model limited to finger radiographs; suggests extending to other limbs and integrating Hadoop.
[10]	Weighted ensemble method combining DenseNet169, MobileNet, and Xception for abnormality detection.	MURA	Ensemble (DenseNet169, MobileNet, Xception)	Higher classification performance than individual networks	Lacks exploration of ensemble diversity; future work may assess scalability and other datasets.
[11]	Introduced MSCNN-GCN: a dual-pathway combining local (MSCNN) and structural (GCN) feature learning.	MURA	MSCNN-GCN	Outperforms conventional CNNs in abnormality classification	Complex architecture; proposed for expansion into lesion segmentation in CT and other modalities.
[12]	Proposed unsupervised deep clustering using ResNet-50 and k-means for anomaly detection across skeletal regions.	MURA	ResNet-50 + k-means Clustering	Effective performance in multi-label anomaly detection	Limited to upper-limb bones; recommends enhancing generalization for device and demographic variation.
[13]	Comparative analysis of YOLOv3, YOLOv7, EfficientDet-D1, and CenterNet for joint localization.	MURA	YOLOv7	mAP@0.5:0.95 = 48.3	Model scaling limited by resources; suggests testing EfficientDet-D7 and hyperparameter tuning.
[14]	Lightweight architecture (FusionNet) combining CA-MBC-EfficientNet-B2 and MobileNetV2 for diagnosis.	MURA	FusionNet	Maintains high diagnostic performance with low computational cost	Requires clinical validation; future work can extend it to wider musculoskeletal regions.

Table 2.1. Methodology Comparison

3. System Architecture and Dataset

The system architecture for self-supervised learning-based abnormality detection in musculoskeletal radiographs consists of a sequential pipeline designed to efficiently learn image representations without heavy reliance on labeled data. The process begins with dataset collection, where the MURA dataset—comprising radiographic images of seven upper limb regions labeled as normal or abnormal—is gathered. This is followed by dataset preprocessing, in which the images are resized to a uniform resolution (e.g., 224×224 pixels), normalized, and converted to a consistent input format suitable for deep learning models. After preprocessing, data augmentation techniques such as random cropping, flipping, color jittering, and Gaussian blur are applied to create multiple views of each image, which are essential for self-supervised learning models to learn robust and invariant features.

The next stage is model training using self-supervised learning (SSL) methods, specifically SimCLR, MoCo, and BYOL, each utilizing a ResNet-50 backbone as the feature encoder. These models are trained to learn meaningful image representations by comparing different augmented views of the same or different images, without requiring labeled data. Once the SSL models have been pretrained, model fine-tuning is conducted using a labeled subset of the dataset. This step adapts the learned representations to the downstream task of abnormality detection either through linear probing or end-to-end training. Finally, the system performs classification, where the fine-tuned model predicts whether a given radiograph is normal or abnormal. The performance is evaluated using standard metrics such as accuracy, F1-score, and AUROC, providing a comprehensive assessment of the model's ability to detect abnormalities in musculoskeletal radiographs.

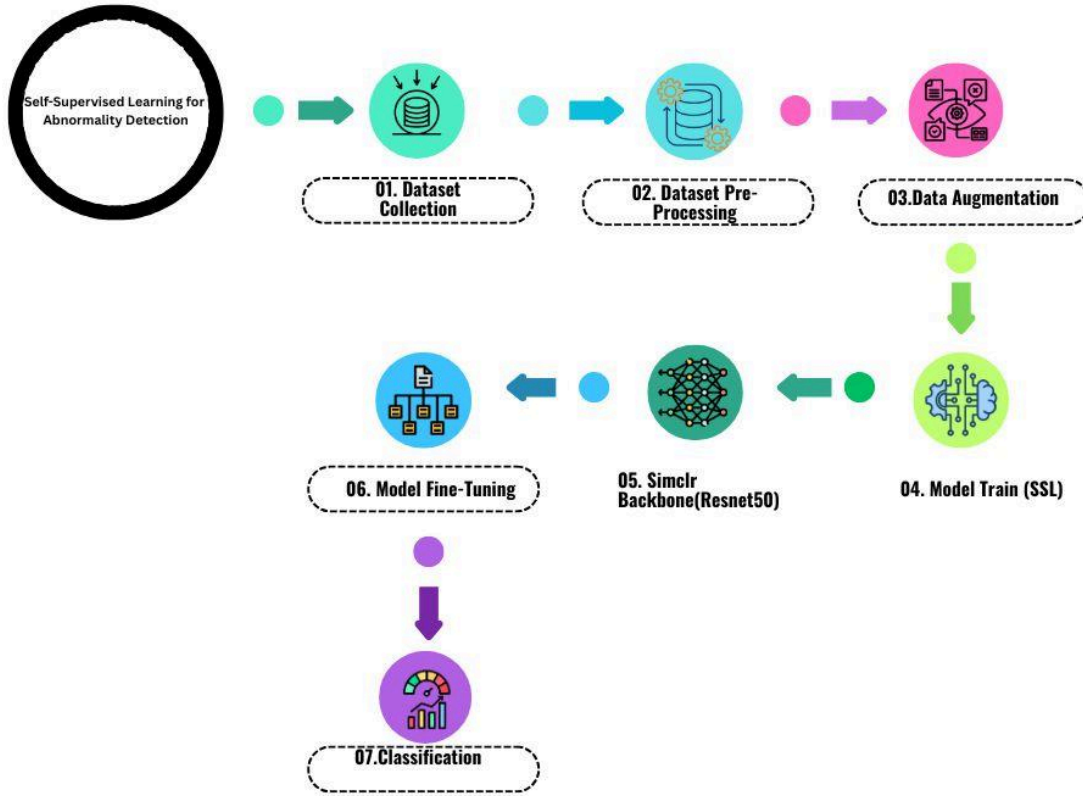


Fig 3.1 System Architecture

3.1. Dataset Description

For this study, we utilized the MURA (Musculoskeletal Radiographs) dataset, a comprehensive collection of musculoskeletal radiographic images released by Stanford ML Group. MURA contains 40,561 images from 14,863 studies of 12,173 patients. The dataset covers seven anatomical regions of the upper extremities: elbow, finger, forearm, hand, humerus, shoulder, and wrist. Each study is labeled by radiologists as either “normal” or “abnormal,” based on the presence or absence of musculoskeletal abnormalities. This dataset poses several challenges due to its class imbalance, inter-class variability, and subtle visual differences in abnormalities. For the purpose of this work, we adopted a binary classification task focusing on detecting abnormalities in any of the regions. We leveraged the full set of radiographs to ensure diversity in image types and improve the generalizability of the learned representations across different anatomical areas.

4.Methodology

4.1 Experimental Setup

System Configuration: For this project, we have used the windows operating system for all the experiment with system configuration of ryzen 7 7700x (16 CPUs), 4.5GHz Up to 5.4GHz, installed RAM of 32 GB 6000 MHz Clock Speed, NVIDIA GeForce RTX 4060 TI 8GB GPU.

Tools Used: We have executed all the experiments using Python on the VSCode code editor with importing different python-based libraries.

4.2 Dataset Pre-Processing

Prior to training, we applied a series of preprocessing steps to ensure consistency and improve the quality of the input data. All radiographs were standardized by resizing them to a resolution of 224×224 times pixels to match the input dimensions expected by the ResNet-50 backbone used in all self-supervised models. Given the grayscale nature of X-ray images, we retained single-channel inputs but replicated them across three channels to maintain compatibility with pre-existing model architectures. Pixel values were normalized to zero mean and unit variance based on the dataset statistics. For self-supervised learning, extensive data augmentation strategies were employed to encourage the models to learn robust and invariant features. These augmentations included random cropping, horizontal flipping, rotation, Gaussian blur, and color jittering, with specific transformations tuned for each model based on their respective learning paradigms.

4.3 Self-Supervised Learning Frameworks

To learn robust visual representations from the unlabeled radiographs, we employed two prominent self-supervised learning methods: SimCLR and MoCo, both using a ResNet-50 backbone as the feature encoder.

SimCLR (Simple Framework for Contrastive Learning of Visual Representations)

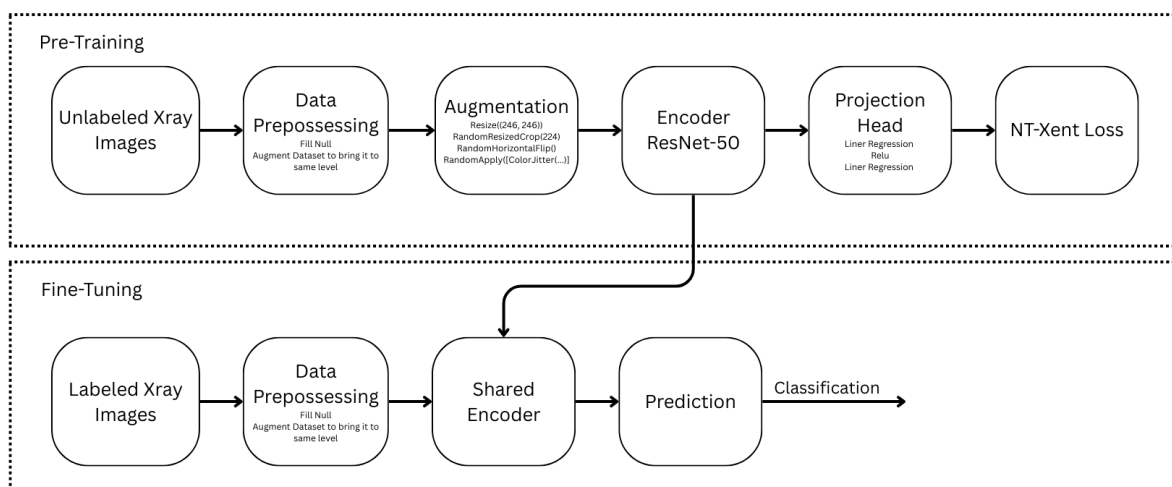
SimCLR is a contrastive learning framework where the objective is to bring different augmented views of the same image (positive pairs) closer in the latent space, while pushing apart views of different images (negative pairs). Each input image undergoes two random augmentations to form a positive pair. These representations are passed through a base encoder (ResNet-50), followed by a projection head—a two-layer multilayer perceptron (MLP) that maps representations to a 128-dimensional latent space. The model is trained using the NT-Xent loss with a temperature parameter of 0.5 to control the sharpness of the distribution. For training SimCLR, we used a batch size of 32, 50 epochs, and an early stopping patience of 5 epochs to prevent overfitting. These settings helped the model learn compact and discriminative features from unlabeled radiographs.

MoCo (Momentum Contrast)

MoCo enhances contrastive learning by maintaining a dynamic queue of negative samples and employing a momentum-based encoder to ensure consistency across training steps. The model uses two encoders: a query encoder (updated via backpropagation) and a key encoder (updated with an exponential moving average of the query encoder). The goal is to match a query representation with its corresponding key while distinguishing it from other negative keys stored in the queue. We implemented MoCo v2 with a ResNet-50 backbone, a projection MLP that maps representations to a 128-dimensional space, and a temperature of 0.07. We used a momentum coefficient of 0.999, a queue size (K) of 4096, and trained the model for 50 epochs using the Adam optimizer with a learning rate of 1e-3. Early stopping was applied with a patience of 5 epochs. This configuration allowed MoCo to learn stable and discriminative features even with smaller batch sizes, thanks to its memory bank mechanism.

4.4 Fine-Tuning and Classification

After pre-training the models using SimCLR and MoCo, we evaluated their effectiveness on the downstream task of abnormality classification. Two evaluation strategies were employed: linear probing and end-to-end fine-tuning. In linear probing, the pretrained encoder was frozen, and a logistic regression classifier was trained on top of the learned features using labeled data. For fine-tuning, the entire network, including the encoder and a new classification head, was trained on the labeled portion of the MURA dataset. This step allowed the models to adapt their learned representations to the specific nuances of medical image classification. Performance was assessed using standard metrics such as accuracy, AUROC, precision, recall, and F1-score.



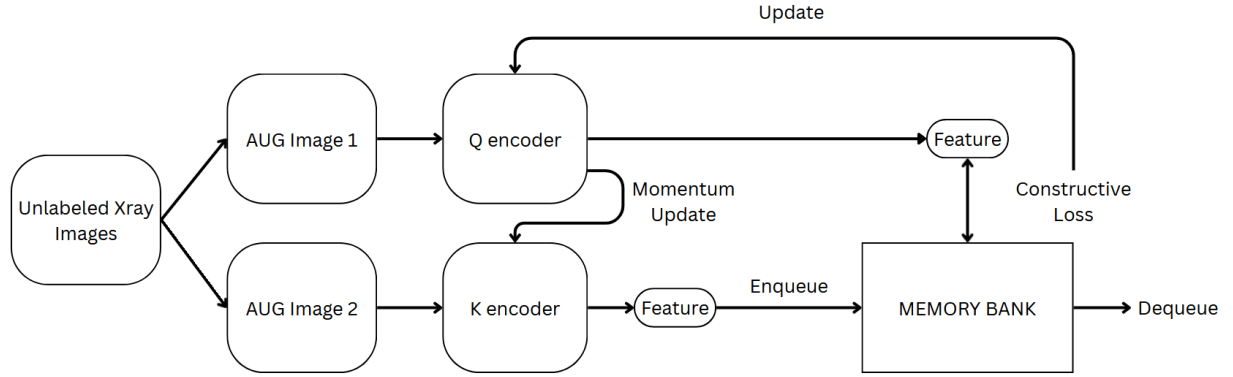


Fig 4.1 Architectural Overview of SimCLR and MoCo for Self-Supervised Representation Learning

4.4 Experimental Results

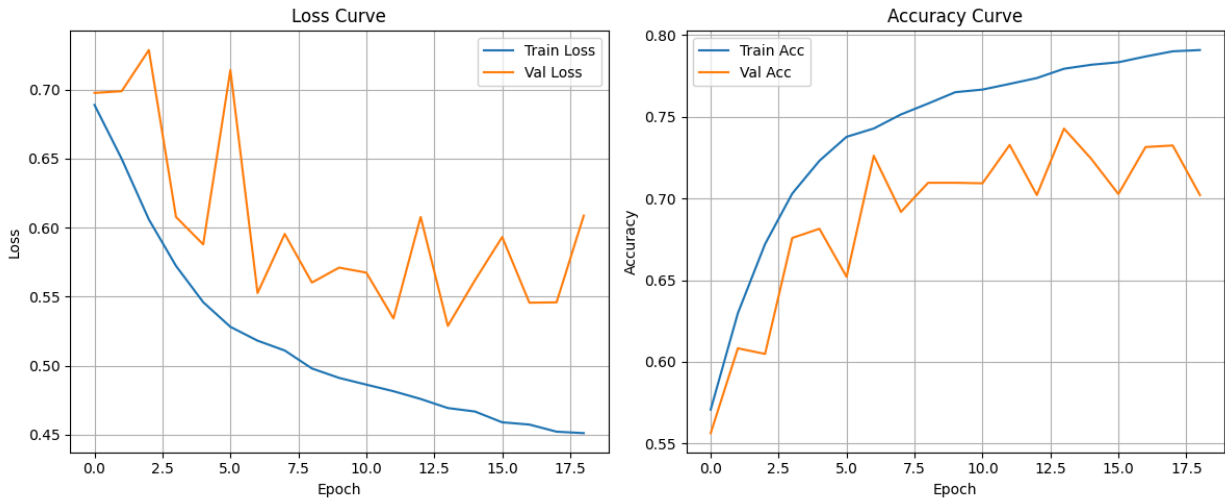


Fig 4.2 Accuracy and Loss Curve Vs Epoch Comparison for SimCLR

The performance of the SimCLR model during pre-training is illustrated in Figure 4.2, which shows the loss and accuracy curves across epochs. The training loss exhibits a steady decline, indicating effective optimization of the contrastive objective and consistent learning of visual representations from the unlabeled radiographs. In contrast, the validation loss fluctuates across epochs, which is typical in contrastive learning settings due to the dynamic nature of sample pairings and batch compositions. The

training accuracy increases progressively, reaching close to 80%, demonstrating that the model effectively distinguishes between positive and negative pairs. Validation accuracy improves in the initial epochs and then stabilizes around 70–72%, suggesting that the learned representations generalize reasonably well to unseen data. While there is a noticeable gap between training and validation accuracy, it remains within acceptable bounds, indicating that the model is not severely overfitting. Overall, these results highlight the effectiveness of SimCLR in learning discriminative and transferable features from the MURA dataset, laying a strong foundation for downstream classification tasks.

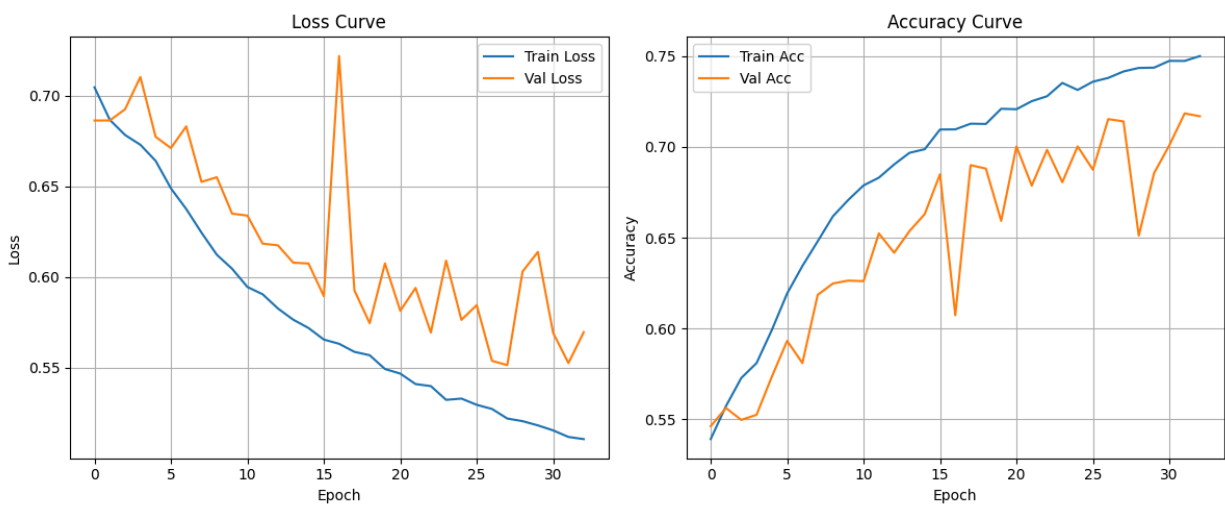


Fig 4.3 Accuracy and Loss Curve Vs Epoch Comparison for MOCO

Figure 4.3 illustrates the training and validation loss and accuracy curves over epochs for the MoCo-based model. The loss curve shows a steady decline in training loss, indicating effective learning, while the validation loss exhibits some fluctuations, suggesting moderate overfitting or sensitivity to data variability. Despite this, the accuracy curve demonstrates consistent improvement in both training and validation accuracy across epochs. The training accuracy steadily increases, reaching approximately 75%, while the validation accuracy stabilizes around 70%, reflecting the model's ability to generalize reasonably well. Overall, the results indicate that the MoCo framework successfully learned meaningful representations that translated into good performance on the classification task.

Model	Training Accuracy	Validation Accuracy	Training Loss	Validation Loss
Sim-CLR	0.8173	0.7725	0.4031	0.4916
MOCO	0.75	0.7168	0.5107	0.5696

Table 4.1 Result Analysis Comparison Among the Models

Table 4.1 presents a comparative analysis of two self-supervised learning models, Sim-CLR and MOCO, evaluated on the task of abnormality detection using the MURA dataset. The performance metrics indicate that Sim-CLR outperforms MOCO across all aspects, achieving higher training accuracy (0.8173) and validation accuracy (0.7725), along with lower training loss (0.4031) and validation loss (0.4916). In contrast, MOCO attained lower training and validation accuracies (0.75 and 0.7168, respectively) and higher losses. These results suggest that Sim-CLR is more effective in learning meaningful representations from the musculoskeletal radiographs and demonstrates better generalization to unseen validation data. Thus, Sim-CLR appears to be a more suitable choice for self-supervised abnormality detection in this context.

5.Result Analysis

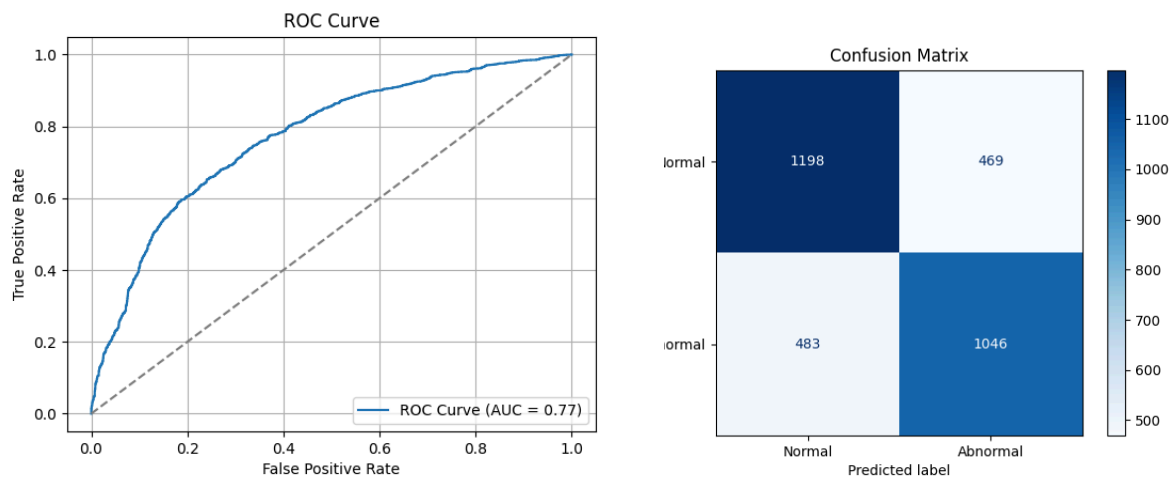


Figure 5.1 ROC Curve and Confusion Matrix (SimCLR)

The ROC curve and confusion matrix presented for the SimCLR-based model provide insight into its classification performance on the abnormality detection task. The ROC curve shows an Area Under the Curve (AUC) of 0.77, indicating a good level of discrimination between normal and abnormal cases. While not perfect, this value reflects the model's reasonable ability to balance sensitivity and specificity across different thresholds. The confusion matrix further supports this evaluation, with 1,046 true positives (correctly identified abnormal cases) and 1,198 true negatives (correctly identified normal cases). However, there are also 469 false positives and 483 false negatives, highlighting areas where the model misclassified samples. These results suggest that while the SimCLR model captures useful representations for the task, there is still room for improvement, particularly in reducing misclassifications to enhance reliability in clinical settings.

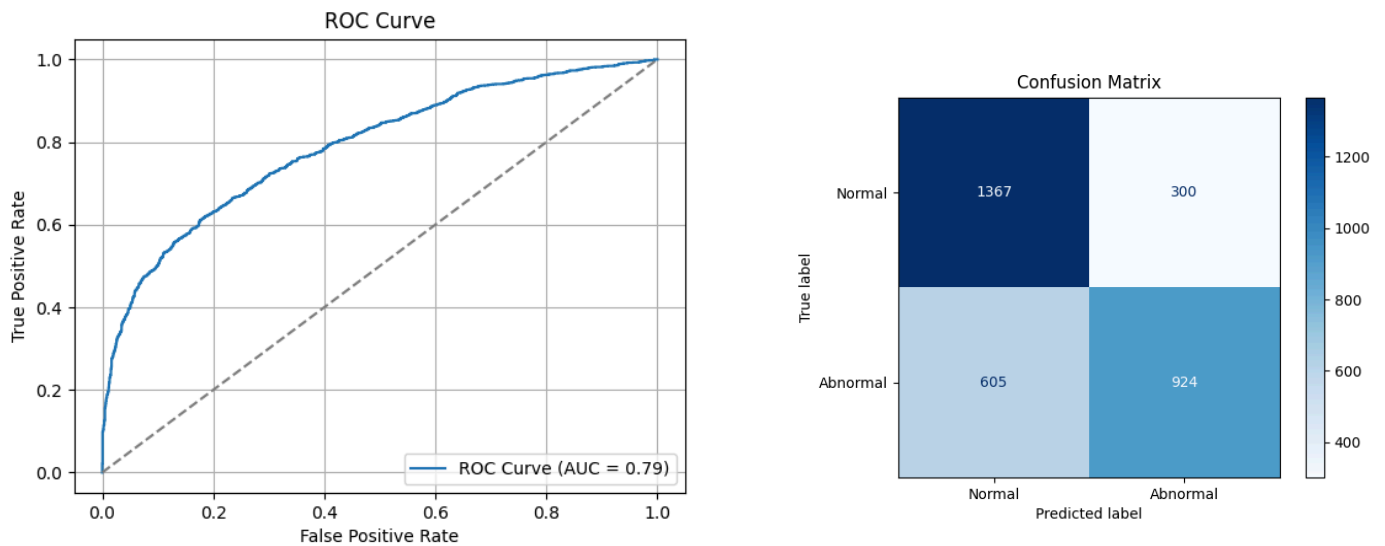
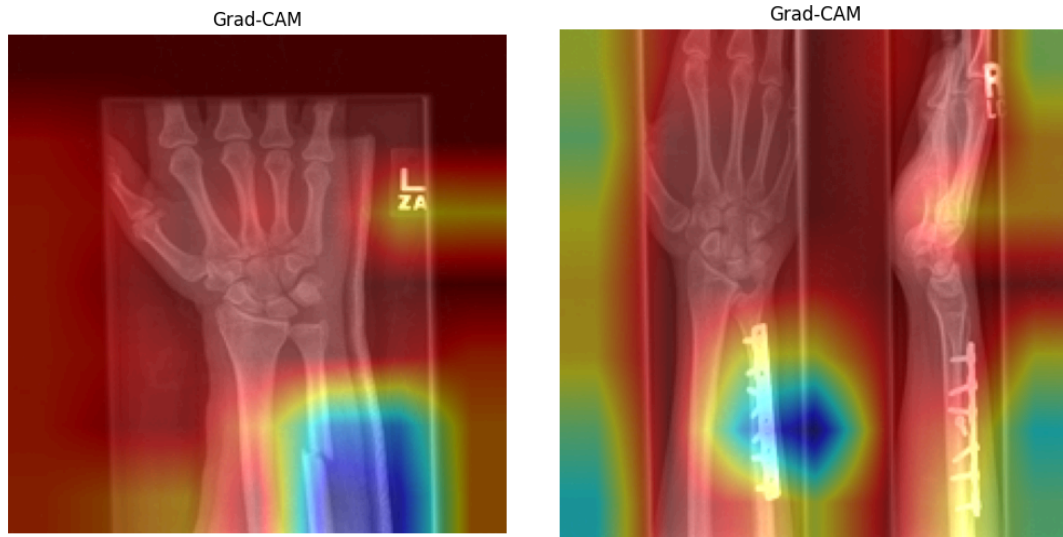


Figure 5.2 ROC Curve and Confusion Matrix (MOCO)

Figure 5.2 presents the ROC curve and confusion matrix for the MoCo-based model. The ROC curve achieves an AUC of 0.79, indicating a strong ability to differentiate between normal and abnormal radiographs. This value reflects an improvement over the SimCLR model, suggesting more robust feature learning through MoCo’s momentum contrast mechanism. The confusion matrix reveals that the model correctly classified 1,367 normal and 924 abnormal cases. However, it also produced 300 false positives and 605 false negatives, with the higher number of false negatives suggesting that some abnormal cases were missed. Despite this, the overall performance indicates that MoCo effectively captures discriminative patterns in the radiographs and is well-suited for downstream classification tasks in a medical imaging context.



Moco

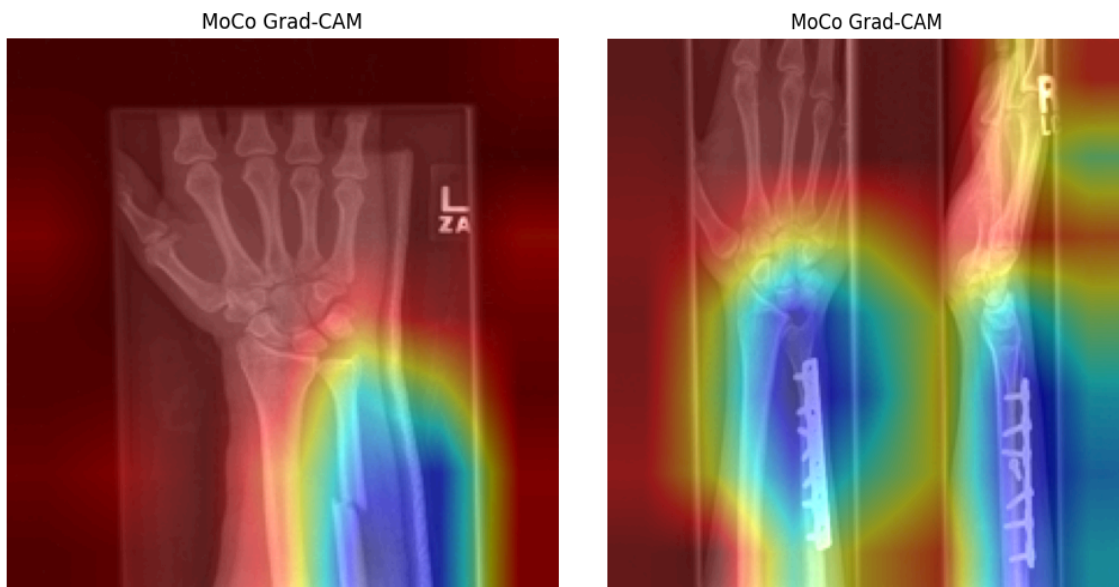


Fig 5.3 Grad-CAM Visualization

The Grad-CAM visualizations presented in Figures 5.3 illustrate the regions of musculoskeletal radiographs that were most influential in the decision-making process of the SimCLR and MoCo models. For SimCLR, the attention maps highlight anatomically relevant regions such as joint spaces and areas with structural abnormalities, indicating that the model successfully learned to focus on clinically meaningful features. Similarly, the MoCo-based Grad-CAM results demonstrate high activation around abnormal bone regions and fracture sites, with more concentrated attention around hardware implants and injury zones. These visualizations validate the interpretability and diagnostic relevance of both self-supervised learning frameworks, showcasing their potential in localizing abnormalities in X-ray images without requiring extensive manual annotations.

5.1.Future Work

While the present study demonstrates the potential of self-supervised learning (SSL) techniques in musculoskeletal radiographic classification, several avenues remain for future exploration and improvement. One promising direction is the incorporation of advanced architectures, such as Vision Transformers and hybrid CNN-transformer models, which may enhance the ability to capture global contextual dependencies in radiographic images. Additionally, expanding the framework to support multi-view and multi-modal learning—by incorporating multiple radiographic views per patient and integrating complementary imaging modalities such as CT and MRI—could significantly improve diagnostic accuracy and generalization in diverse clinical scenarios.

Another critical area involves the adoption of semi-supervised and active learning paradigms. These approaches can optimize annotation efficiency by identifying and labeling the most informative samples, thereby further reducing dependence on large labeled datasets. Equally important is the enhancement of model interpretability through the integration of explainable AI techniques such as SHAP, and LIME. These tools can provide visual rationales for model predictions, fostering transparency and building trust among clinical practitioners.

Real-world clinical validation represents a vital step in translating SSL models into practical diagnostic tools. This involves rigorous testing in clinical environments, including prospective studies and interactive evaluations with radiologists, to assess performance, usability, and workflow integration. Finally, to improve model robustness and fairness, future research should address limitations within the dataset—such as class imbalance, limited demographic representation, and imaging variability. Expanding the dataset to include a more diverse patient population and a broader range of anatomical variations will be essential to ensure generalizability and reliability across real-world settings.

By addressing these areas, future research can advance the clinical applicability and reliability of self-supervised learning approaches in musculoskeletal radiograph analysis, paving the way for more intelligent and accessible diagnostic systems.

5.2. Conclusion

This study investigated the efficacy of self-supervised learning (SSL) frameworks—specifically SimCLR and MoCo—for abnormality detection in musculoskeletal radiographs using the MURA dataset. By leveraging contrastive learning mechanisms, both models demonstrated the ability to learn robust and transferable visual representations from unlabeled radiographic images, significantly reducing the dependency on large labeled datasets.

Quantitative evaluation revealed that the MoCo-based model achieved superior classification performance, with an AUC of 0.79 compared to 0.77 for SimCLR, highlighting the effectiveness of momentum contrastive learning in capturing meaningful features. While both models achieved satisfactory performance in terms of accuracy and AUROC, the presence of false positives and false negatives underscores the need for further refinement before clinical deployment.

Overall, the findings confirm that SSL represents a promising paradigm for building scalable, label-efficient computer-aided diagnosis systems in the musculoskeletal imaging domain. These methods not only mitigate the challenges posed by limited annotated data but also offer a strong foundation for developing advanced diagnostic tools that can augment radiologist decision-making.

Reference:

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